Sinusoidal Signals

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Key Concepts

1) Sinusoids play an important role in signal processing.

    a) They occur frequently in physical phenomenon.

    b) We are familiar with the notion of frequency in sound, seasons, and other periodic phenomena.

    c) Any physical signal can be represented as a weighted sum of sinusoids.

2) Discrete- and continuous-time sinusoids are described by an amplitude, frequency, and phase.

3) The frequency of a sinusoid is the inverse of the period.

   a) Period is measured in the number of samples (discrete-time) or seconds (continuous-time) for one "cycle" of the sinusoid.

      i) One cycle corresponds to the angle of a continuous-time sinusoid changing by 2π radians.

      ii) One cycle corresponds to the angle of a discrete-time sinusoid changing by $2\pi m$ radians where m is integer.

   b) Frequency of a continuous-time sinusoid is measured in cycles/second (Hertz - Hz), or radians/second.

c) Frequency of a discrete-time sinusoid is measured in radians/sample or radians.

4) Continuous-time sinusoids with distinct frequencies are always distinct.

5) Discrete-time sinusoids with frequencies shifted by integer multiples of 2π are identical.

6) Sampling relates the frequency Ω rads/sec of a continuous-time sinusoid to the frequency ω rads of a discrete time sinusoid as $\omega = \Omega T$ where T is the sampling interval.

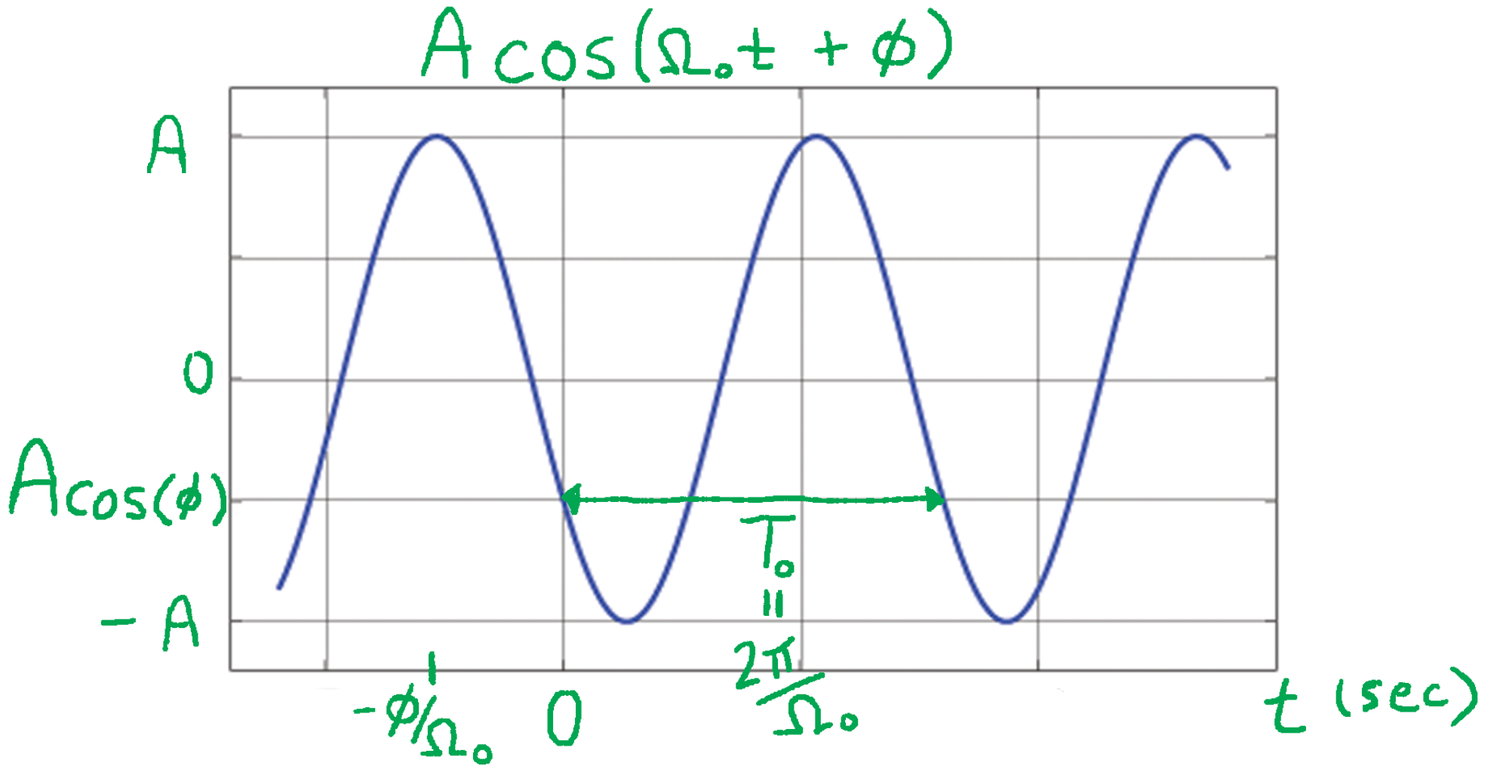
Video Screenshots

Why Sinusoids?

1. Occur in nature
    a. light of a given color
    b. microwaves
    c. oscillatory motion: pendulum, undamped spring-mass system
2. Notions of "frequency" pervade our experience
3. Bases: sums of sinusoids can describe any signal

Continuous-Time Sinusoids

$$x(t) = A\cos(\Omega_o t + \phi)$$

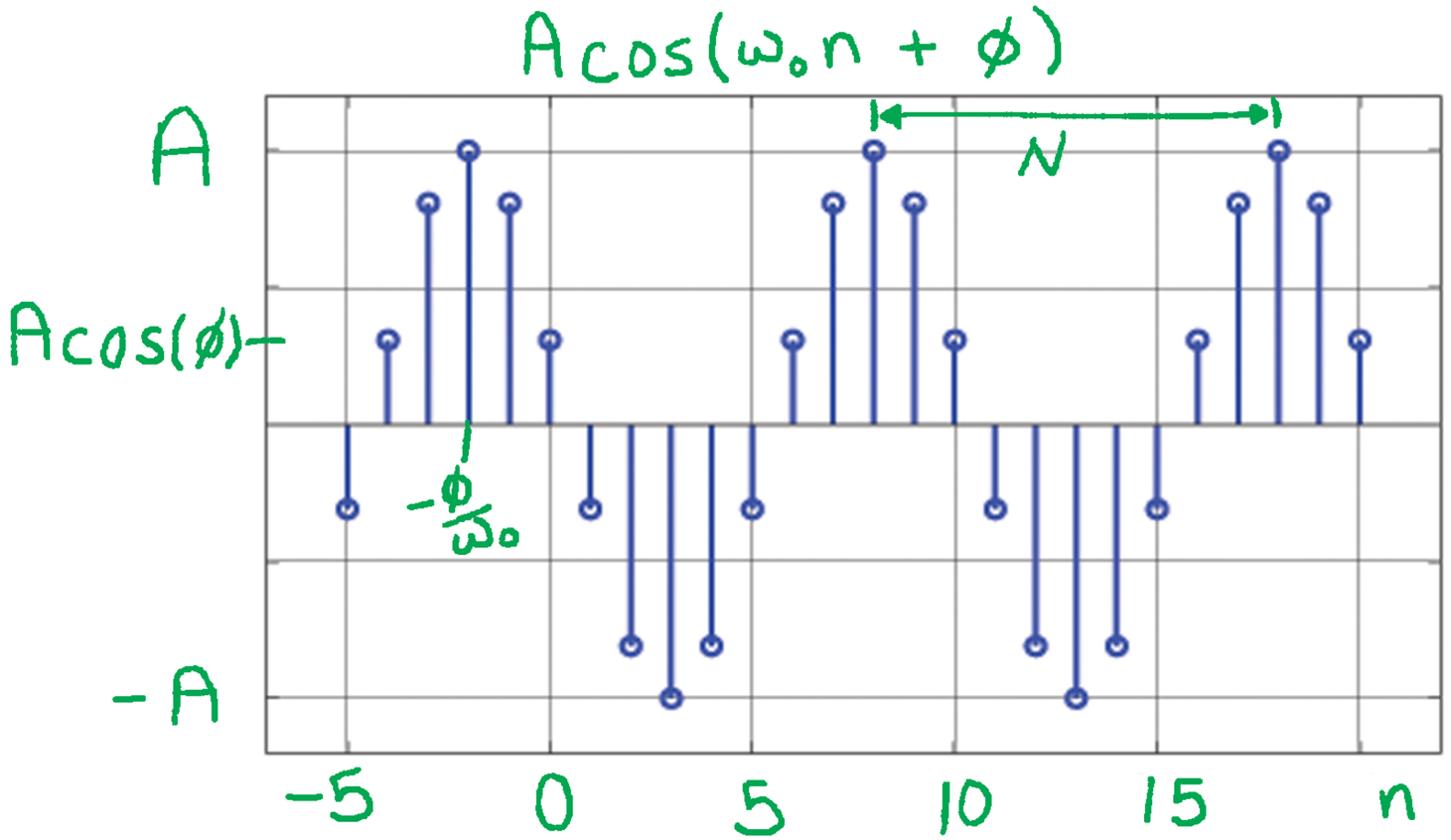


Amplitude: A
Phase Shift: ϕ
Frequency: Ω_o

Period: $T_o = \dfrac{2\pi}{\Omega_o}$

Discrete-Time Sinusoids

$$x[n] = A\cos(\omega_o n + \phi)$$

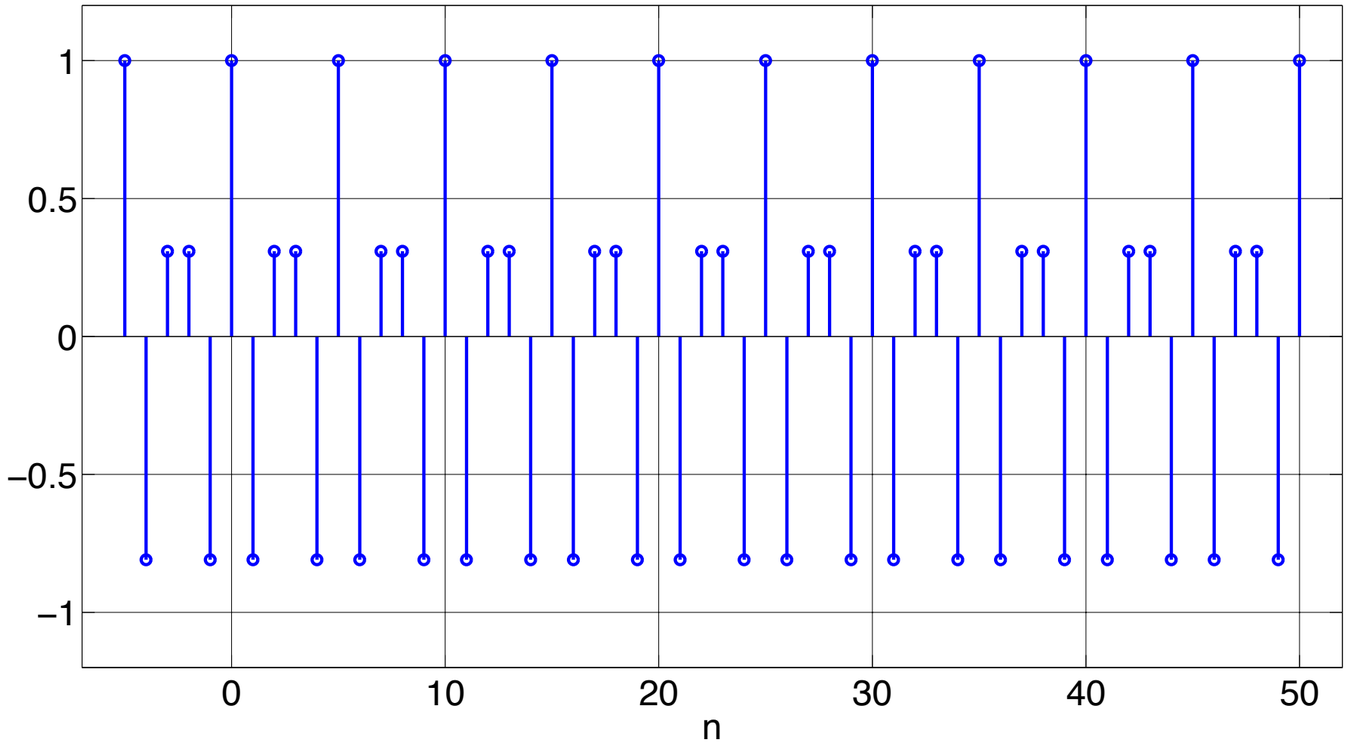


Amplitude: A
Phase Shift: ϕ
Frequency: ω_o

Period: N

Discrete-Time Sinusoid Periodicity

$$x[n] = \cos\left(\frac{4\pi}{5}n\right)$$



$$x[n] = A\cos(\omega_o n + \phi)$$

$N,\ n$ Integers

$$\omega_o = \frac{2\pi m}{N} \quad \text{integer } m$$

Frequency

- Frequency = 1/Period
- Period measured in seconds (CT) or samples (DT)
- Frequency measured in radians/time or cycles/time

CT Period: T_o sec

$$f_o = \frac{1}{T_o} \text{ cycles/sec or Hz}$$

$$2\pi \text{ rads/cycle}$$

$$\Omega_o = 2\pi f_o = \frac{2\pi}{T_o} \text{ rads/sec}$$

DT Period: N samples

$$\frac{1}{N} \text{ cycles/sample}$$

$$2\pi m \text{ rads/cycle}$$

$$\omega_o = \frac{2\pi m}{N} \text{ rads/sample or rads}$$

Relating Continuous- and Discrete-Time Frequency

$$x[n] = A\cos(\omega n + \phi) \quad \textbf{vs} \quad x(t) = A\cos(\Omega t + \phi)$$

- Force samples of CT sinusoid to equal DT sinusoid

$$\omega = \Omega T$$

$$x[n] = x(t)|_{t=nT}$$

$$\begin{aligned} A\cos(\omega n + \phi) &= A\cos(\Omega n T + \phi) \\ &= A\cos\left((\Omega T)n + \phi\right) \end{aligned}$$

$$\text{rads} = \frac{\text{rads}}{\text{sec}} \times \text{sec}$$

$$\omega = 2\pi f T$$

$$\text{rads} = \frac{\text{rads}}{\text{cycle}} \times \frac{\text{cycles}}{\text{sec}} \times \text{sec}$$

Uniqueness

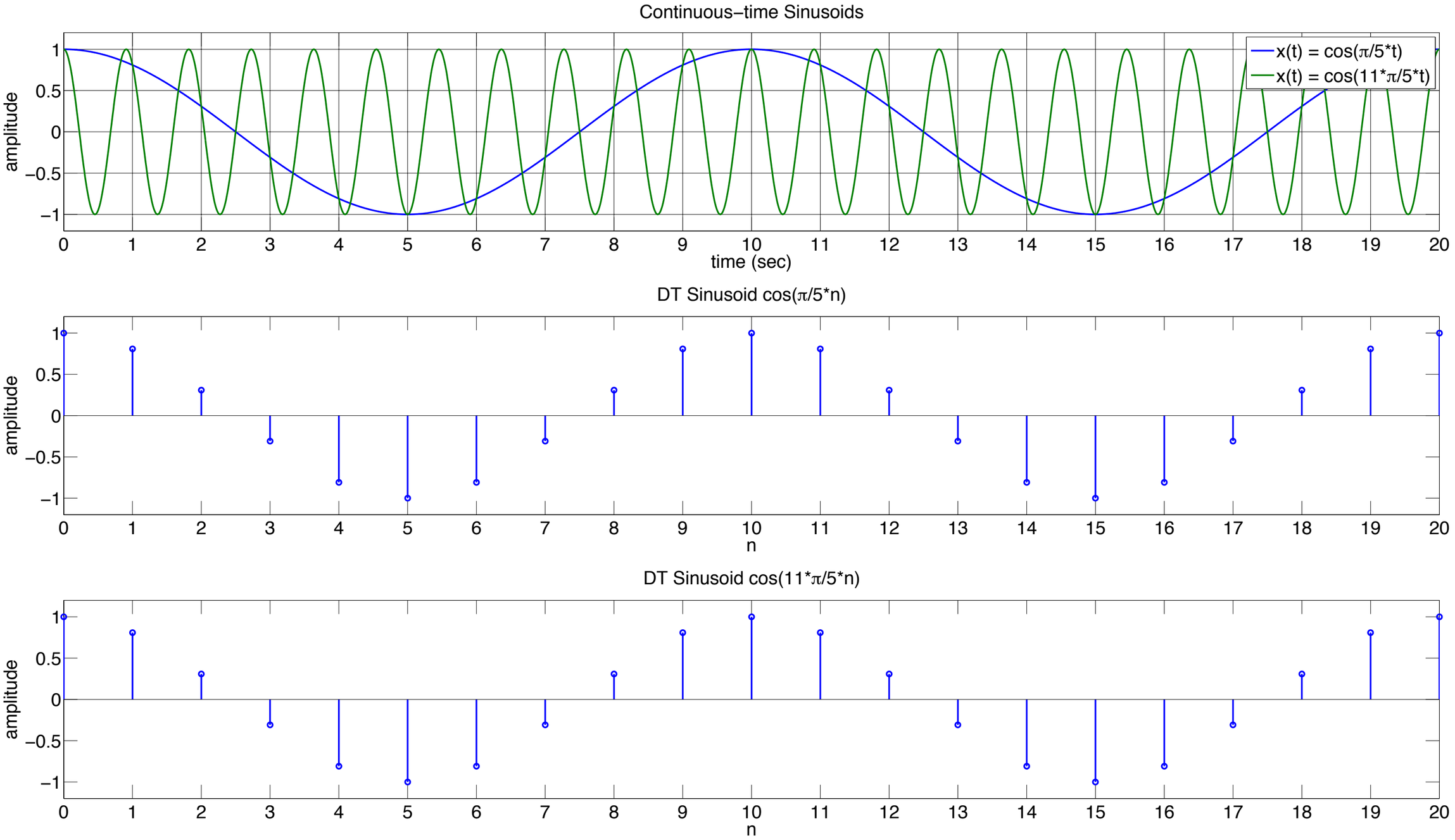
- Continuous-time: different frequencies → different sinusoids

$$\text{If } \Omega_1 \neq \Omega_2, \text{ then } \sin(\Omega_1 t) \neq \sin(\Omega_2 t)$$

- Discrete-time: shift frequency by $m2\pi$ → same sinusoid

$$\text{If } \omega_2 = \omega_1 + m2\pi, \text{ then } \sin(\omega_2 n + \phi) = \sin(\omega_1 n + \phi)$$

$$\begin{aligned}
\sin(\omega_2 n + \phi) &= \sin((\omega_1 + m2\pi)n + \phi) \\
&= \sin(\omega_1 n + \phi)\cos(m2\pi n) + \cos(\omega_1 n + \phi)\sin(m2\pi n) \\
&= \sin(\omega_1 n + \phi)
\end{aligned}$$


Continuous−time Sinusoids
x(t) = cos(π/5*t)
x(t) = cos(11*π/5*t)
amplitude
time (sec)
DT Sinusoid cos(π/5*n)
amplitude
n
DT Sinusoid cos(11*π/5*n)
amplitude
n

Summary - Sinusoids

- Frequency is inversely related to the period
- Discrete- and continuous-time frequency are related through sampling

$$\omega = \Omega T$$

- Discrete-time sinusoids are only unique over $0 \leq \omega < 2\pi$

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